

2-2-3 Cryostat Cooling / Cryogen Filling Requirements

- Record the Diode Temperatures. See SET-UP AND CALIBRATION, Section 2-2-1.

J301U Diode 1 _____

J301L Diode 1 _____

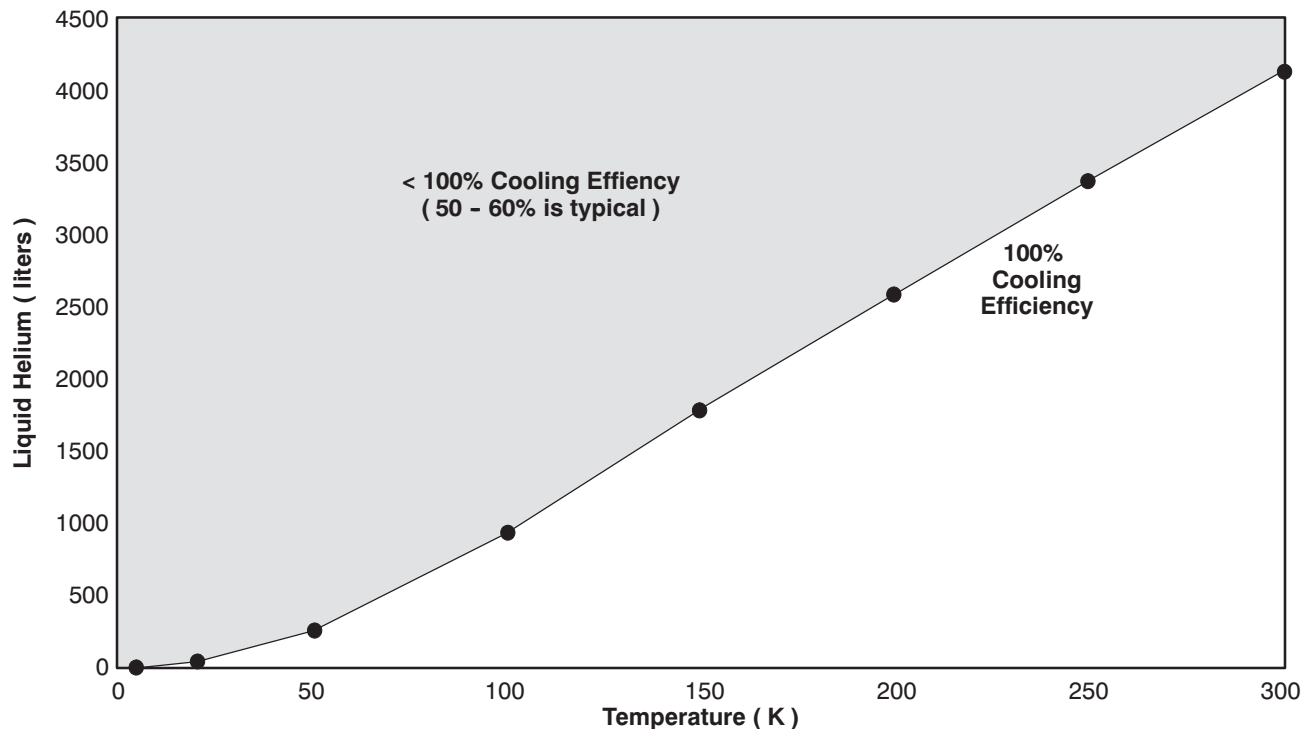


Magnet temperature checks are performed during installation to determine the Cryostat cooling / cryogen filling requirements prior to ramping. Make sure the indicated procedures, based upon diode temperatures, are followed.

- Determine the approximate liquid volume required using Illustration 2-5 as a guideline. Divide the liquid helium volume by an efficiency factor typical for your equipment and process to determine the actual liquid helium volume required.

Note

The line in Illustration 2-5 represents 100% efficient helium cooling. A typical range for transfer / cooling efficiency is 50 - 60% and is dependent on a number of process and equipment factors.



**LIQUID HELIUM REQUIRED TO COOLDOWN TO 4.2K,
AS A FUNCTION OF INITIAL MAGNET TEMPERATURE**
ILLUSTRATION 2-5